

# **BASIC CONCEPTS OF NETWORK**



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# DEFINITION & APPLICATIONS

## □ DEFINITION:

**A computer network is defined as the interconnection of two or more computers. It is done to enable the computers to communicate and share available resources.**

## □ APPLICATIONS:

- i. Sharing of resources such as printers**
- ii. Sharing of expensive software's and database**
- iii. Communication from one computer to another computer**
- iv. Exchange of data and information among users via network**
- v. Sharing of information over geographically wide areas.**

# COMPONENTS OF COMPUTER NETWORK

- **Two or more computers**
- **Cables as links between the computers**
- **A network interfacing card(NIC) on each computer**
- **Switches**
- **Software called operating system(OS)**

# NETWORK BENEFITS

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- **The network provided to the users can be divided into two categories:**
  - i. **Sharing**
  - ii. **Connectivity**

# SHARING RESOURCES

## □ **Types of resources are:**

- 1. Hardware: A network allows users to share many hardware devices such as printers, modems, fax machines, CD ROM, players, etc.**
- 2. Software: sharing software resources reduces the cost of software installation, saves space on hard disk.**

# OTHER BENEFITS OF COMPUTER NETWORK

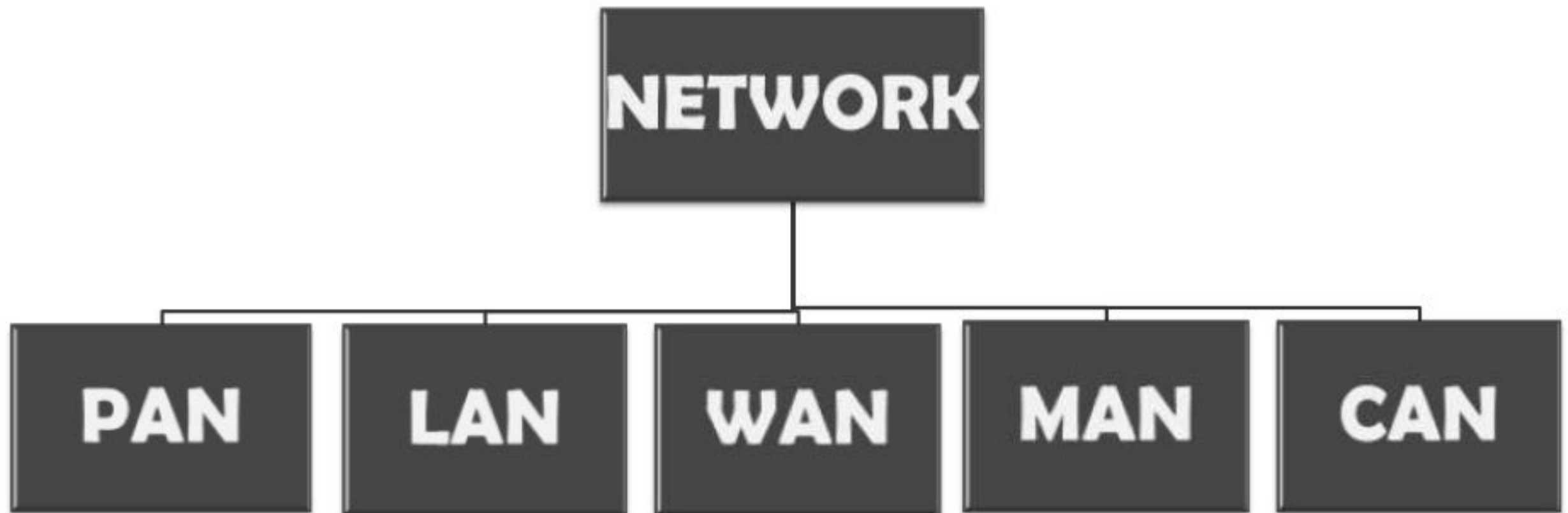
- **Increased speed**
- **Reduced cost**
- **Improved security**
- **Centralized software managements**
- **Electronic mail**
- **Flexible access**

# DISDAVATAGES OF NETWORKS

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- **High cost of installation**
- **Requires time for administration**
- **Failure of server**
- **Cable faults**

# CLASSIFICATION OF AREA BY THEIR GEOGRAPHY



# LOCAL AREA NETWORK(LAN)

- ❑ **LAN is a network which is designed to operate over a small physical area such as an office, factory or a group of buildings.**
- ❑ **LAN's are easy to design and troubleshoot**
- ❑ **Exchange of information and sharing of resources becomes easy because of LAN.**
- ❑ **In LAN all machines are connected to a single cable.**
- ❑ **Different types of topologies such as star, tree, bus, ring, etc Can be used**
- ❑ **It is usually a privately owned network.**

# WIDE AREA NETWORK(WAN)

- **When network spans over a large distance or when the computers to be connected to each other are at widely separated locations a local area network cannot be used. A wide area network(WAN) is installed.**
- **The communication between different users of WAN is established using leased telephone lines, satellite links and similar channels.**
- **It is cheaper and more efficient to use the phone network for the link.**
- **Most WAN networks are used to transfer large blocks of data between its users.**

# PERSONAL AREA NETWORK(PAN)

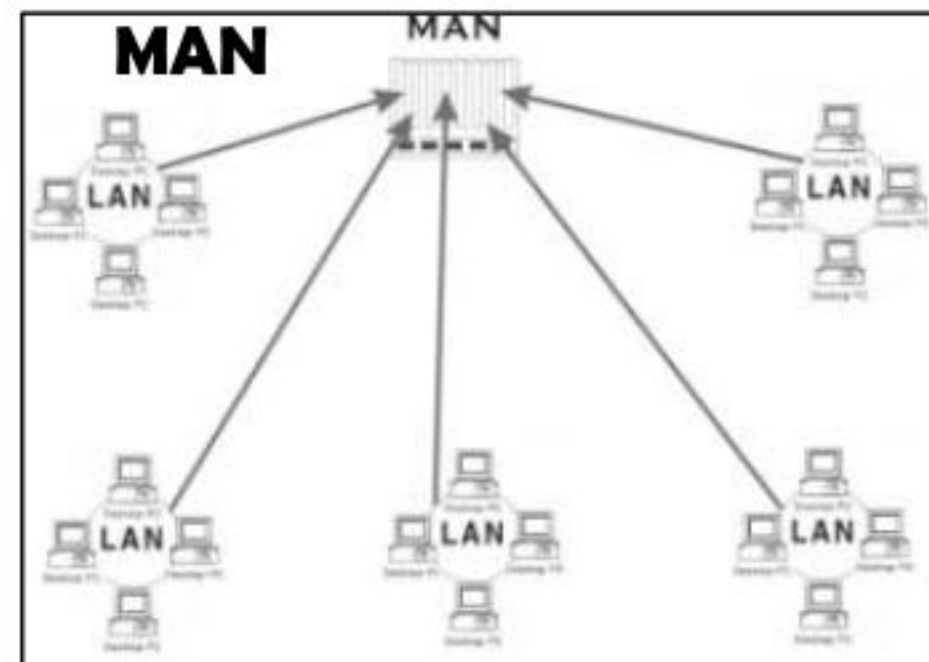
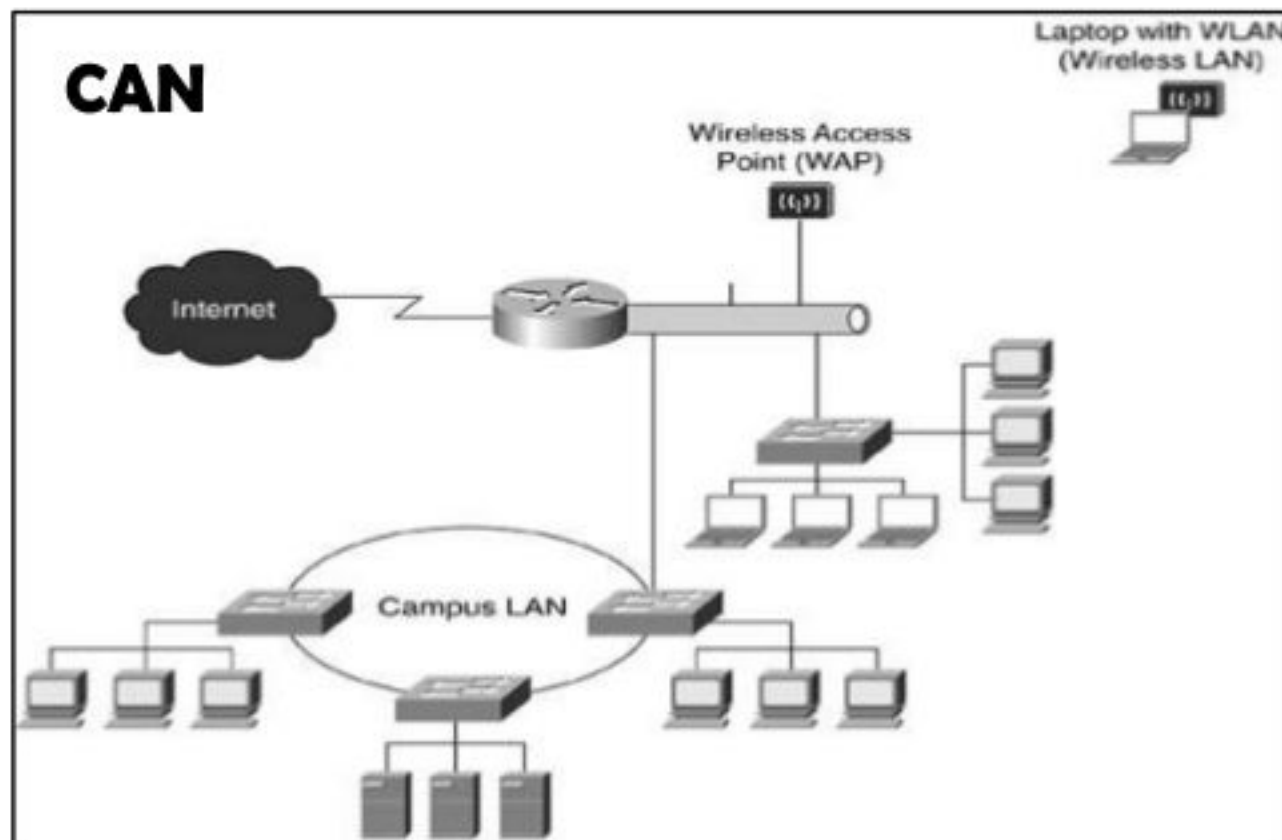
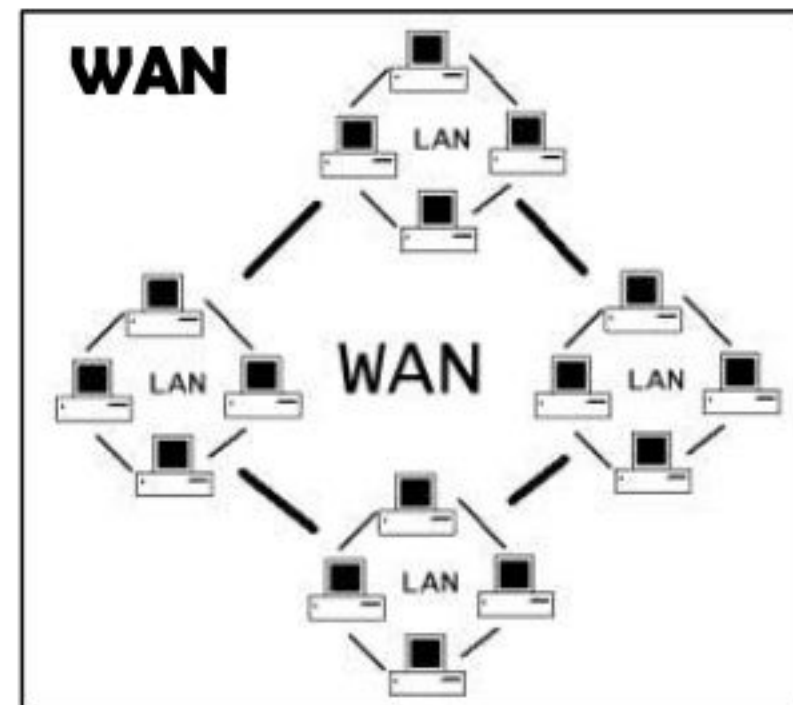
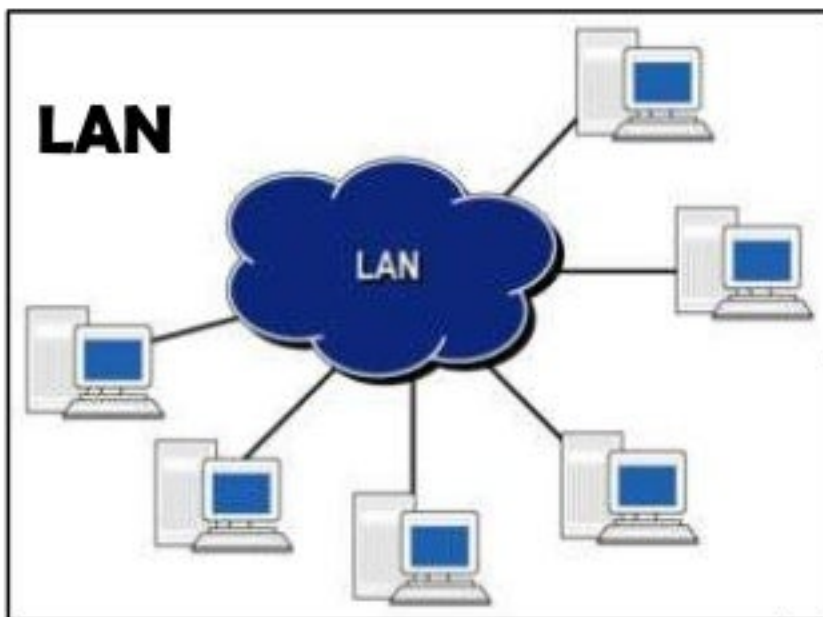
- ❑ **A personal area network is a computer network organized around an individual person.**
- ❑ **It generally consists of a mobile computer, a cell phone or personal digital assistant. PAN enables the communication among these devices.**
- ❑ **It can also be used for communication among personal devices themselves for connecting to a digital level network and internet.**
- ❑ **The PANs can be constructed using wireless or cables.**

# CAMPUS AREA NETWORK(CAN)

- ❑ **The campus area network is made up of an interconnection of LAN with limited geographical area.**
- ❑ **Network equipments such as switches, routers and the transmission media i.e. optical fibre etc are almost entirely owned by the campus owner.**

# METROPOLITAN AREA NETWORK(MAN)

- ❑ **It is in between LAN & WAN technology that covers the entire city.**
- ❑ **It uses similar technology as LAN.**
- ❑ **It can be a single network such as cable TV network, or a measure of connecting a number of LAN's o a large network so that resources can be shared LAN to LAN as well as device to device.**

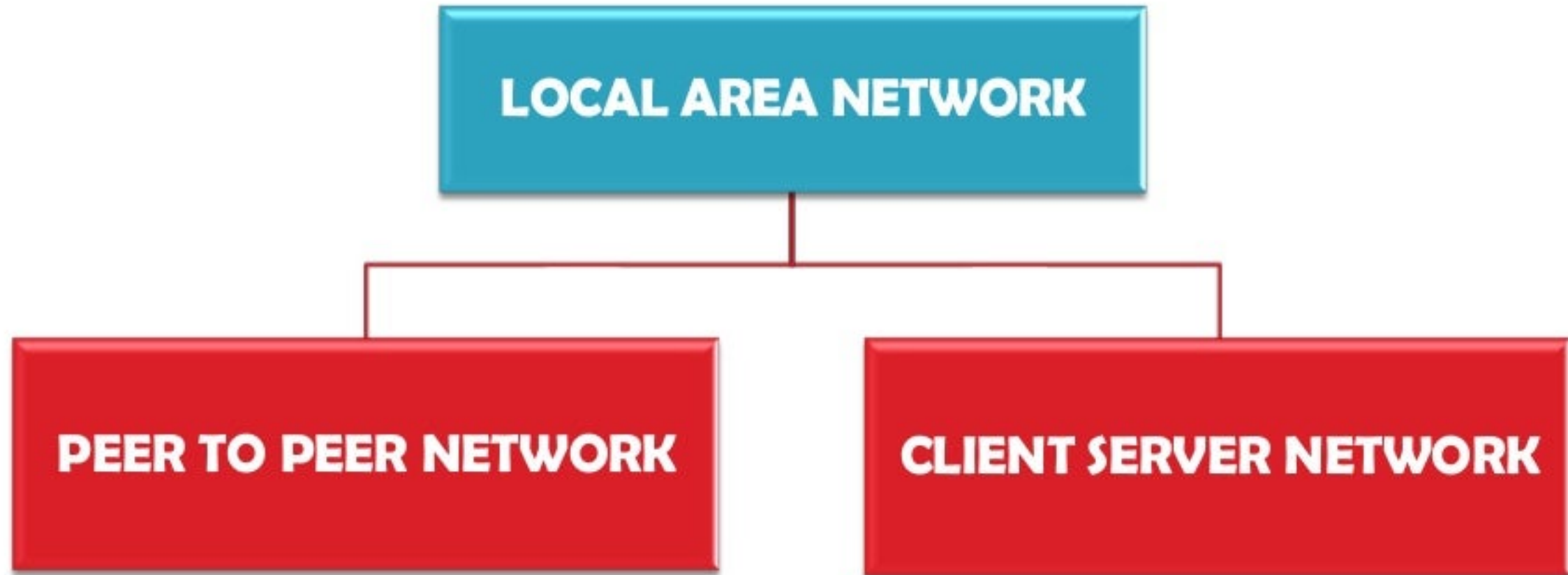


# DISTINGUISH BETWEEN LAN,WAN,MAN

| PARAMETERS                | LAN           | WAN                     | MAN                                                   |
|---------------------------|---------------|-------------------------|-------------------------------------------------------|
| Ownership of network      | Private       | Private or public       | Private or public                                     |
| Geographical area covered | Small         | Very large              | Moderate                                              |
| Design and maintenance    | Easy          | Not easy                | Not easy                                              |
| Communication medium      | Coaxial cable | PSTN or satellite links | Coaxial cables, PSTN, optical fibre, cables, wireless |
| Bandwidth                 | Low           | High                    | moderate                                              |
| Data rates(speed)         | High          | Low                     | moderate                                              |

# **NETWORK CLASSIFICATION BY THEIR COMPONENT ROLE**

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# PEER TO PEER NETWORK

- ❑ **In peer to peer network each computer is responsible for making its own resources available to other computers on the network.**
- ❑ **Each computer is responsible for setting up and maintaining its own security for these resources.**
- ❑ **Also each computer is responsible for accessing the required network resources from peer to peer relationships.**
- ❑ **Peer to peer network is useful for a small network containing less than 10 computers on a single LAN .**
- ❑ **In peer to peer network each computer can function as both client and server.**
- ❑ **Peer to peer networks do not have a central control system. There are no servers in peer networks.**
- ❑ **Peer networks are amplified into home group.**

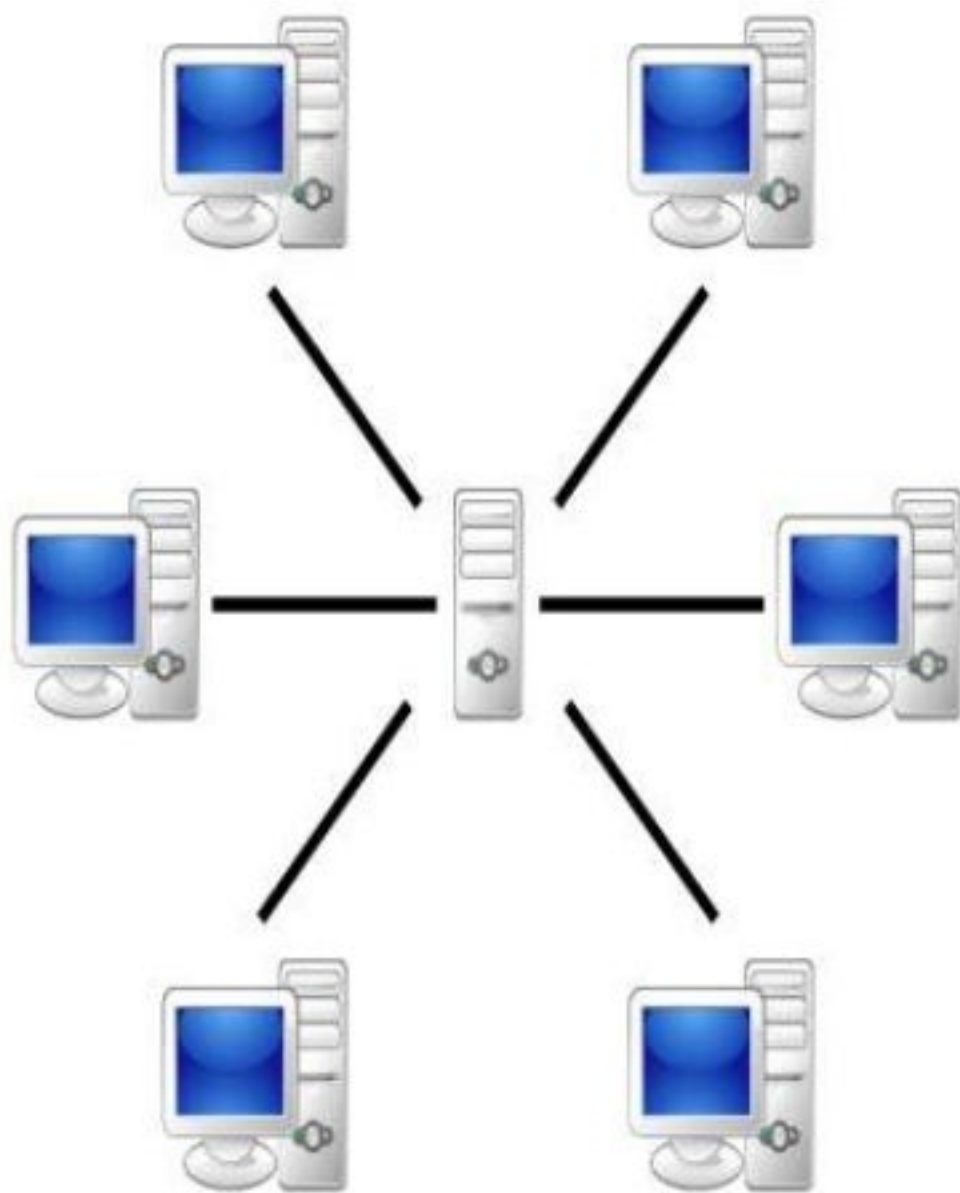
# ADVANTAGES & DISADVANTAGES OF PEER TO PEER NETWORK

## Advantages:

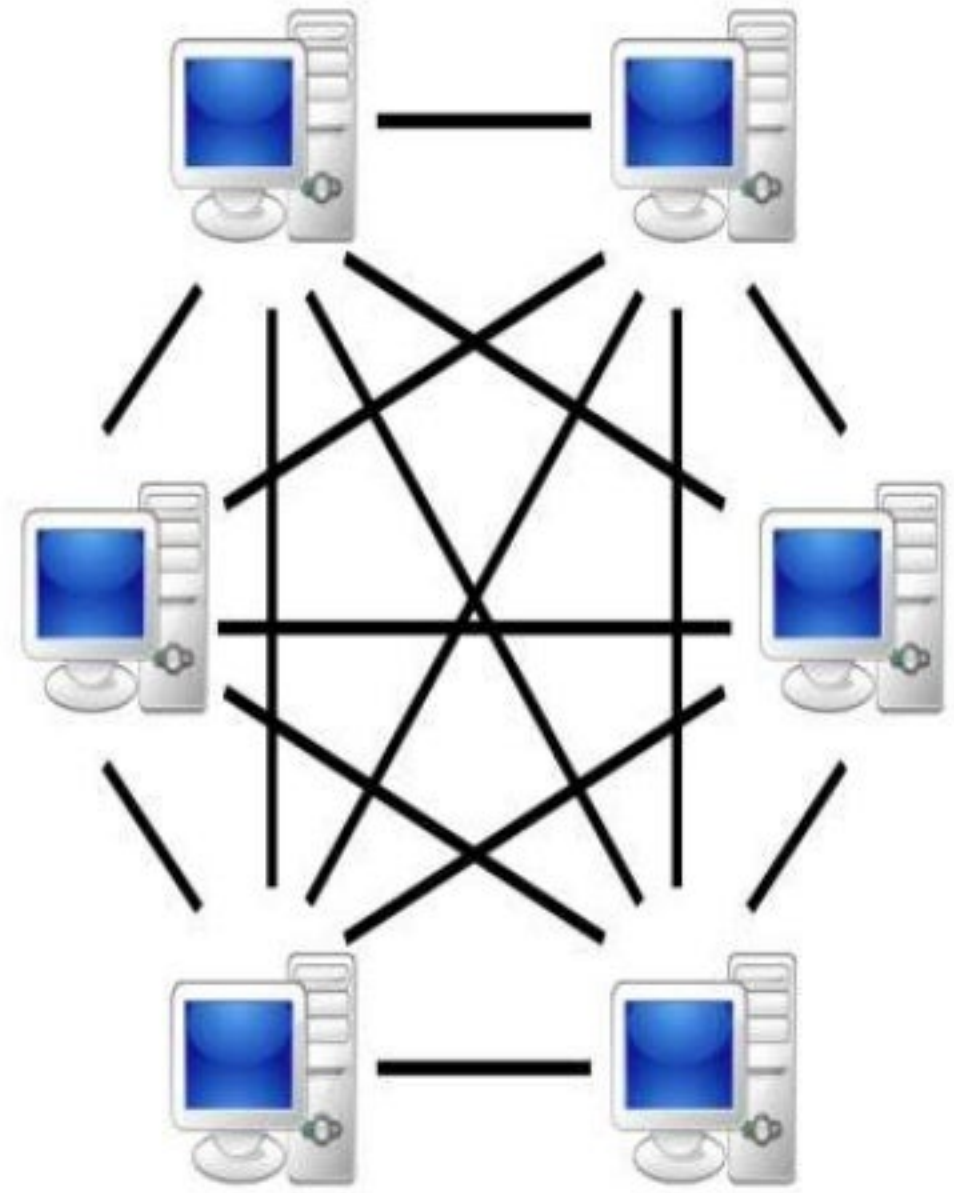
- **Use less expensive computer hardware**
- **Easy to administer**
- **No NOS required**
- **More built in redundancy**
- **Easy setup & low cost**

## Disadvantages:

- **Not very secure**
- **No central point of storage or file archiving**
- **Additional load on computer because of resource sharing**
- **Hard to maintain version control**



Server-based



P2P-network

# CLIENT/SERVER NETWORK

- ❑ In client-server network relationships, certain computers act as server and other act as clients. A server is simply a computer, that available the network resources and provides service to other computers when they request it. A client is the computer running a program that requests the service from a server.
- ❑ Local area network(LAN) is based on client server network relationship.
- ❑ A client-server network is one in which all available network resources such as files, directories, applications and shared devices, are centrally managed and hosted and then are accessed by client.
- ❑ Client server network are defined by the presence of servers on a network that provide security and administration of the network.

# ADVANTAGES AND DISADVANTAGES OF CLIENT-SERVER NETWORK

## Advantages:

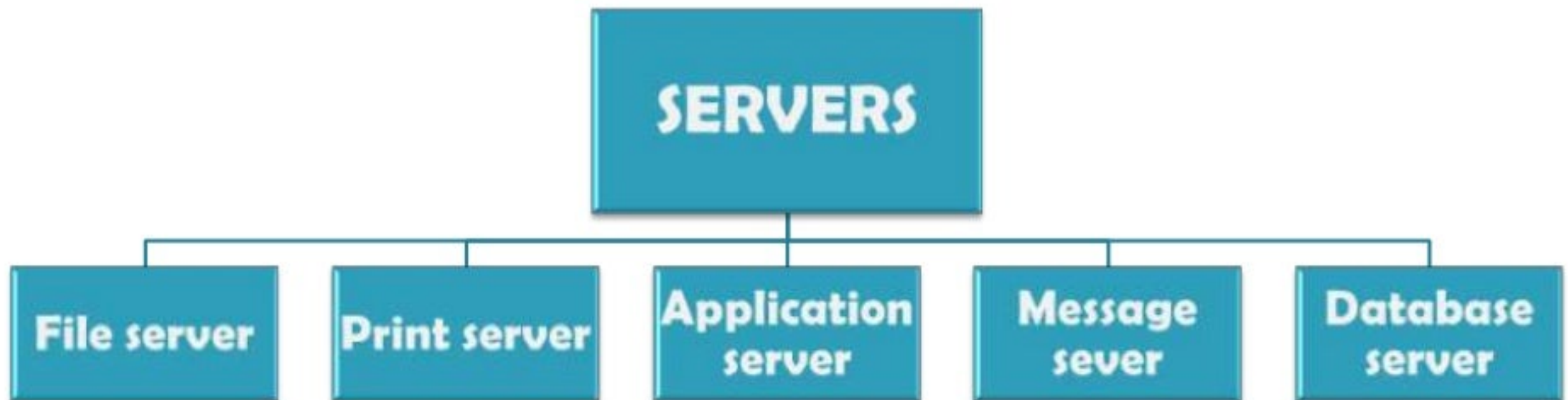
- **Very secure**
- **Better performance**
- **Centralized backup**
- **very reliable**

## Disadvantages:

- **requires professional administration**
- **More hardware-intensive**
- **More software intensive**
- **Expensive dedicated software**

# TYPES OF SERVERS

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# TYPES OF SERVERS

- ❑ **File server:** These servers provide the services for storing, retrieving and moving the data. A user can read, write, exchange and manage the files with the help of file servers.
- ❑ **Printer server:** The printer server is used for controlling and managing printing on the network. It also offers the fax service to the network users.
- ❑ **Application server:** The expensive software and additional computing power can be shared by the computers in a network with the help of application servers.
- ❑ **Message server:** It is used to co-ordinate the interaction between users, documents and applications. The data can be used in the form of audio, video, binary, text or graphics.
- ❑ **Database server:** It is a type of application server. It allows the users to access the centralised strong database.